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<https://doi.org/10.63720/vii2007>**Coronary Artery Bypass or Percutaneous Intervention in Chronic Coronary Syndrome: A Focused Review of the 2024 European Society of Cardiology Guidelines**Elena Yuen-Yee Lee^{1*} , Chun Aun Koo¹, Yi Lun Khaw¹, Ujjawal Kumar² , Harry James Smith³ , Fadi Al-Zubaidi⁴ , and Daniel Sitaranjan⁵ **Abstract**

This review summarises the 2024 ESC guidelines on coronary revascularisation in chronic coronary syndromes (CCS), comparing the roles of CABG and PCI across various clinical scenarios. Based on the best available evidence, the recommendations are defined as class I (should be performed), IIa (reasonable to perform), IIb (may be considered), and III (should not be performed). CABG remains the preferred strategy for left main and complex multivessel disease, particularly in patients with diabetes mellitus, owing to its superior long-term outcomes. PCI offers comparable survival in non-diabetic patients with low SYNTAX scores and preserved ventricular function. For single- or double-vessel disease involving the proximal left anterior descending artery (LAD), both PCI or CABG provide prognostic benefit, while isolated non-LAD disease may be managed with PCI for symptoms not controlled by medical therapy. In ischemic heart failure patients (ejection fraction $\leq 35\%$) with viable myocardium, CABG improves survival, whereas PCI is an alternative if the surgical risk is prohibitive. Overall, the guidelines emphasise that revascularisation decisions should be individualised through a multidisciplinary Heart Team approach integrating evidence, available expertise, and patient preferences.

Key Words: Chronic Coronary Syndromes (CCS), Myocardial Revascularisation, Percutaneous Coronary Intervention (PCI), Coronary Artery Bypass Grafting (CABG), ESC 2024 Guidelines, SYNTAX Score, Heart Team

Introduction

Chronic coronary syndromes (CCS) encompass patients with stable coronary artery disease who may experience angina or myocardial infarction in the long term.¹ In these patients, optimal management includes guideline-directed medical therapy and, for appropriate cases, myocardial revascularisation using percutaneous coronary intervention (PCI) or coronary artery bypass grafting (CABG).²

The 2024 European Society of Cardiology (ESC) Guidelines for CCS provide detailed recommendations on when to pursue revascularisation and how to choose between PCI and CABG.³ These recommendations are based on large clinical trials and meta-analyses, and are graded by Class (I, IIa, IIb, III) based on Levels of Evidence (A, B, and C) for clinical decision-making.³ A patient-centred, multidisciplinary “Heart Team” approach is essential to deliver the most appropriate treatment and hence is necessitated by the latest guidelines.⁴

Herein, we present a clinician-oriented summary comparing PCI vs CABG according to the 2024 ESC CCS guidelines, including the rationale behind recommendations, indications in various clinical scenarios, and key outcomes that guide modality selection.

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Indications for Revascularisation in Chronic Coronary Syndromes

In CCS, revascularisation is generally indicated to improve survival in certain high-risk anatomic subsets and to relieve angina in patients who remain symptomatic despite medical therapy.⁴⁻⁷ The guidelines underscore that significant left main coronary artery disease ($\geq 50\%$ stenosis) or extensive multivessel disease (three-vessel disease with $\geq 70\%$ stenoses, or two-vessel disease including proximal LAD stenosis) are key indications where revascularisation offers prognostic benefit. In these scenarios, especially for left main and/or proximal LAD involvement, revascularisation (by either CABG or PCI as appropriate) is recommended over medical therapy alone to improve survival and clinical outcomes (Class I).⁸

Revascularisation is also indicated for symptom control in patients with angina that is not adequately controlled by optimal medical therapy.^{7,9} For example, in a CCS patient with severe angina and a functionally significant lesion in the proximal LAD (even single-vessel), an intervention (PCI or CABG) may be recommend-

ed to improve symptoms and prevent adverse events (Class I).⁸ The choice between CABG and PCI depends on multiple factors: the extent and complexity of coronary disease (often quantified by SYNTAX score), the presence of comorbidities (especially diabetes mellitus or left ventricular dysfunction), patient surgical risk and preferences, and the likelihood of achieving complete revascularisation with each method.^{2,10-12} Generally, CABG is favoured in more diffuse or complex disease and in patients who have high-risk features (e.g. diabetes or impaired LV function) because of its more durable results and survival benefit.^{2,13-15} PCI is often favoured in patients with limited disease complexity or in those who are poor surgical candidates, owing to PCI's less invasive nature and faster recovery.^{2,11,16} Importantly, the guidelines stress that decisions should be individualised via a Heart Team consultation, particularly when both PCI and CABG seem equivalent and/or carry the same class of recommendation.^{17,18} The Heart Team—comprising interventional cardiologists, cardiothoracic surgeons, non-invasive cardiologists, and others—should weigh anatomy, patient comorbidities, life expectancy, and patient preferences to select the optimal strategy.^{19,20} Shared decision-making with the patient is paramount, ensuring the patient's values and choices are factored into the revascularisation plan.²⁰

Comparative Evidence-Based: PCI vs CABG Outcomes

The ESC recommendations draw on evidence from multiple randomised trials and meta-analyses comparing PCI and CABG in stable CAD. Overall, current evidence indicates that CABG tends to confer a longer-term survival advantage in certain populations and reduces spontaneous myocardial infarctions (MI) and repeat revascularisation compared to PCI. For example, in patients with complex multivessel disease (especially those with diabetes), trials like FREEDOM and a number of meta-analyses have shown that CABG significantly lowers rates of death and MI over follow-up relative to PCI, at the cost of a higher upfront procedural risk.²¹⁻²³ CABG's benefit is partly attributed to more complete revascularisation (including bypassing of multiple lesions and protecting downstream myocardium) and the longevity of grafts, which translates to fewer repeat procedures.^{10,24-25} Indeed, the guidelines note that among patients anatomically eligible for both strategies, repeat re-intervention is consistently needed more often after PCI than after CABG, even with current-generation drug-eluting stents.²⁶⁻²⁷

By contrast, PCI offers the advantage of lower invasiveness and faster recovery, which can be important for patient preference and peri-procedural safety in high-risk surgical patients. In subsets of patients with less complex coronary anatomy, PCI has demonstrated outcomes comparable to CABG. In patients with left main or three-vessel disease of low anatomical complexity (e.g. SYNTAX score ≤ 22), contemporary trials have shown no significant difference in all-cause mortality at five years between PCI (with drug-eluting stents) and CABG, while PCI circumvent the morbidity of surgery.^{10,28-29} The guidelines explicitly state that in such low-complexity scenarios where complete revascularisation can be achieved percutaneously, PCI's survival is

non-inferior to CABG. However, even in these cases, differences persist in non-mortality outcomes: CABG tends to better prevent spontaneous (non-procedural) MI, whereas PCI carries a higher need for subsequent revascularisation (often due to disease progression or in-stent restenosis).^{21,30,31} Both modalities have a similar long-term stroke risk, though CABG is known to have a higher peri-procedural stroke incidence.³² These nuances in outcomes underscore why the guidelines tailor recommendations to patient subgroups—maximising survival and MI reduction (favouring CABG) versus minimising invasiveness and preserving comparable outcomes in select cases (favouring PCI).

Another critical piece of evidence is the presence of diabetes mellitus. Data (e.g. the FREEDOM trial and others) have consistently shown that diabetic patients with multivessel disease derive a greater survival and MI reduction benefit from CABG than from PCI.¹³ Conversely, PCI outcomes in diabetics have been less favourable, likely due to more diffuse atherosclerosis and higher risk of stent failure.^{13,33,34} This evidence is reflected in stronger recommendations for CABG in diabetics and a caution or even contraindication for PCI in certain scenarios (detailed below). Similarly, in patients with ischemic cardiomyopathy (e.g. left ventricular ejection fraction $\leq 35\%$ due to CAD), surgical revascularisation (as shown by trials like STICH) improves survival versus medical therapy, whereas the benefit of PCI in this setting is less established.^{14,15} The guidelines incorporate these findings by preferentially recommending CABG when significant viable—but ischemic—myocardium exists in HFrEF patients.

The best available evidence can be distilled as follows: surgical revascularisation is more complete and durable than PCI, with proven survival benefit in extensive disease (especially with diabetes or impaired LV function), while PCI yields comparable survival in lower-risk anatomy and offers advantages in less invasiveness and initial safety. These principles inform the guideline's class-of-recommendation for each clinical scenario that will be summarised in the next paragraphs.

Guideline Recommendations by Clinical Scenario

Left Main Coronary Artery Disease

For CCS patients with significant left main (LM) coronary stenosis, the guidelines strongly favour revascularisation for prognostic benefit. In a patient at low surgical risk with $\geq 50\%$ LM stenosis, CABG is recommended over medical therapy alone to improve survival (Class I, Level A).^{35,37} Furthermore, CABG is recommended as the overall preferred revascularisation modality over PCI in left main disease, given CABG's lower risk of spontaneous MI and need for repeat surgical revascularisation (Class I, A).^{2,21,28,37,38} This means that, all else being equal, surgery is the first-line revascularisation strategy for left main CAD in suitable surgical candidates (Table 1).

However, the guidelines also acknowledge that PCI can be an acceptable alternative in select left main patients with less complex disease. If the left main lesion is associated with low anatomical complexity (SYNTAX score ≤ 22) and complete revascularisation can be achieved with PCI, then PCI is recommended as an alter-

Table 1: Left Main Coronary Artery Disease

Clinical Scenario	CABG Recommendation	PCI Recommendation	Key Considerations
Low complexity (SYNTAX ≤ 22)	Class I, Level A (preferred overall)	Class I, Level A (acceptable alternative)	Both equally endorsed if complete revascularization achievable with PCI
Intermediate complexity (SYNTAX 23-32)	Class I, Level A (preferred)	Class IIa, Level A (reasonable to consider)	CABG preferred due to outcome differences at moderate complexity
High complexity (SYNTAX >32)	Class I, Level A (strongly preferred)	Not recommended (by omission)	Surgical revascularization is default for very complex lesions
High surgical risk/ inoperable	Contraindicated (due to risk)	Class IIb, Level B (may be considered with caution)	PCI as compromise when CABG not feasible

native to CABG in those patients (Class I, Level A).⁸ In other words, for a low-SYNTAX left main (for example, isolated left main disease or left main plus one-vessel disease with favourable anatomy), PCI and CABG both carry a Class I/A recommendation—PCI is considered non-inferior to surgery in terms of survival in this subgroup, while being less invasive. The heart team should then weigh patient preference and operative risk in making the final recommendation.

For intermediate anatomical complexity left main disease (SYNTAX 23–32), PCI is given a slightly lower profile: it “reasonable to consider” (Class IIa, Level A) in cases where, again, equivalent revascularisation completeness can be attained percutaneously.^{11,39–41} This represents a downgrade from Class I to IIa for PCI in intermediate-SYNTAX left main, reflecting some lingering outcome differences at moderate complexity in favour of CABG. At high complexity (SYNTAX >32), the implication is that PCI is generally not recommended—the guidelines do not explicitly list a Class III, but by omission and prior evidence, surgical revascularisation is the default for very complex left main lesions.^{2,34,40–42}

It is important to highlight the scenario of the patient with left main disease who is a high surgical risk or inoperable: patients with significant left main CAD who have prohibitive surgical risk (due to age, frailty, comorbidities), the guidelines allow that PCI may be considered over medical therapy alone as a revascularisation strategy (Class IIb, B).^{2,4} This ensures that even those who cannot undergo CABG have an option for revascularisation; PCI in this context is a compromise approach to improve symptoms (and possibly prognosis) when CABG is not feasible. A heart team evaluation is critical here to document the surgical risk and to proceed with PCI only if the associated risk of surgery outweighs the expected benefits.

In summary, for left main CCS: CABG is generally preferred for its outcome benefit (Class I), PCI is equally endorsed if anatomy is favourable (Class I for low complexity, IIa for intermediate), and PCI can be done as the second best option in high-risk surgical patients (IIb).

Notably, the 2024 guidelines were endorsed by both cardiology and cardiothoracic surgery societies, resolving prior debates over left main management by focusing on complete revascularisation and SYNTAX score in guiding PCI vs CABG.⁸

Multivessel Coronary Artery Disease (Three-Vessel CAD)

Three-vessel coronary artery disease (3VD) is another scenario where revascularisation improves the long term outcome, notably, the guidelines differentiate recommendations based on the presence of diabetes and disease complexity (Table 2).

For multivessel disease in patients with diabetes mellitus (DM), the evidence strongly favours surgery. In CCS patients with significant three-vessel CAD and coexistent type 2 diabetes, particularly if angina persists despite commencing on medical therapy according to guidelines, CABG is recommended over both medical therapy alone and over PCI to improve symptoms and outcomes (Class I, Level A).^{13,43} This strong recommendation for CABG in diabetics reflects robust data that CABG reduces mortality and MI in diabetic multivessel patients, whereas PCI is inferior in this group. Accordingly, the guidelines essentially prioritise CABG as the revascularisation method of choice for diabetic patients with multivessel disease. In fact, PCI is not recommended for diabetic 3-vessel patients if the anatomical complexity is intermediate or high (SYNTAX > 22), given the higher event rates observed with stenting in this cohort (Class III, A for PCI in DM with SYNTAX > 22). Only if the anatomy is relatively simple (SYNTAX 0–22) and the surgical risk is not acceptable might PCI be considered, and even then, the guideline assigns this a weak recommendation (approximately Class IIb) for diabetics. Thus, unless there are extenuating circumstances, a diabetic with multivessel disease should be directed to CABG for prognostic benefit. One such extenuating circumstance is very high surgical risk: in a diabetic patient who is an extremely high-risk surgical candidate (e.g. due to severe comorbidities, frailty, reduced life expectancy, etc.), the guidelines state that PCI is reasonable to consider over medical therapy to relieve ischemia and improve short term

Table 2: Three-Vessel Disease

Patient Population	CABG Recommendation	PCI Recommendation	Key Considerations
With diabetes mellitus + acceptable surgical risk	Class I, Level A	Class III, Level A (not recommended if SYNTAX >22)	Robust evidence for CABG superiority in diabetics
Diabetes + very high surgical risk	High risk procedure	Class IIa, Level B (reasonable to consider)	PCI when CABG truly not an option
Without diabetes (preserved LV function)	Class I, Level A	Class I, Level A (if low-intermediate complexity, SYNTAX ≤32)	Equal recommendations when anatomy suitable for PCI
Without diabetes (high complexity)	Class I, Level A (preferred)	Lower priority (not explicitly stated)	Higher Priority for CABG

outcome (Class IIa, B).^{6,44} This provides a path for intervention when CABG is truly not an option, although the expected benefits with PCI would be more for symptom control than for equal longevity benefit in this subgroup.

For three-vessel disease in patients without diabetes (and with preserved LV function), the recommendations are a bit more balanced. If the patient remains symptomatic on medical therapy, revascularisation is recommended over medical therapy alone to improve symptoms, survival, and reduce ischemic events (Class I, A).^{45,47} In other words, revascularisation is clearly indicated for significant 3VD in non-diabetics as well. When it comes to choosing the mode of revascularisation in non-diabetics, the guidelines give a nod to PCI in cases of lower complexity: in a patient with 3-vessel CAD of low-to-intermediate anatomical complexity (again roughly SYNTAX ≤ 32) where PCI is deemed to achieve a similar completeness of revascularisation as CABG, PCI is recommended as an alternative to surgery (Class I, Level A).^{48,52} The wording “PCI is recommended” (Class I) in this context is significant – it effectively places PCI on equal footing with CABG for non-diabetic 3VD when anatomy is favourable. The rationale is that for such patients, studies have shown generally non-inferior survival with PCI versus CABG, especially in the era of improved stents and medical therapy – be it with follow up limited to five years or less. Therefore, the choice can be individualised based on patient preference, with the understanding that PCI will have the advantage of less invasiveness while offering similar hard outcomes in the short-to-medium term.

If the 3-vessel anatomy in a non-diabetic is very complex (e.g. diffuse disease, high SYNTAX) such that complete revascularisation by PCI is doubtful, then CABG remains the preferred modality (Class I). The ESC guidelines incorporate this by restricting the Class I indication for PCI to low-intermediate complexity disease, and by recommending a Heart Team evaluation of anatomy, SYNTAX score, and likelihood of complete revascularisation before choosing PCI over CABG in 3VD.⁸

In practice, for a non-diabetic patient with 3-vessel CAD, if the SYNTAX score is low or moderate and lesions are suitable, PCI can be offered as an equal alternative (with Class I support), whereas if the disease is extensive, CABG should be selected to maximise survival benefits

and freedom from MI.

Summary for 3-vessel disease: For diabetic patients, CABG is strongly favoured (Class I/A), PCI generally discouraged except possibly in low SYNTAX cases (PCI is Class IIb at best, and Class III/not recommended if SYNTAX > 32). For non-diabetic patients, both CABG and PCI are Class I options if anatomy is suitable for PCI; otherwise, CABG is indicated. In all cases, ensure symptoms or ischemia justify revascularisation (since in asymptomatic low-risk disease, one might continue medical therapy). The Heart Team should consider diabetes status, SYNTAX score, anticipated completeness of revascularisation, and comorbidities when deciding on PCI vs CABG in multivessel disease.

Single or Double-Vessel Disease

For patients with one or two-vessel coronary disease, the need for CABG vs PCI depends largely on whether the proximal left anterior descending artery (LAD) is involved and on symptom burden. The proximal LAD supplies a large territory of the myocardium; thus, a significant proximal LAD stenosis carries a higher risk, as with multivessel disease, and thus merits durable revascularisation strategy (Table 3).

In CCS patients with significant single-vessel or two-vessel disease that involves the proximal LAD, if angina and/or ischemia persists despite optimal medical therapy, revascularisation (either PCI or CABG) is recommended over medical therapy alone to improve symptoms and outcomes (Class I, A).⁸ This means that even if a patient has otherwise limited disease, a high-grade proximal LAD lesion should be fixed (assuming the patient is a suitable candidate), because doing so improves prognosis and quality of life.^{45,54–57} In practice, PCI is often the first choice for an isolated proximal LAD lesion given its accessibility and the patient's likely preference to avoid surgery; however, CABG using the left internal mammary artery in particular, is an equally valid approach, particularly if the LAD lesion is not easily amenable to stenting (e.g. very long or calcified plaque). It is worth noting that in patients with complex lesions in the LAD, there is a higher chance that repeated revascularisation at a later stage would be required after PCI compared to CABG.⁵⁸ In the meantime, due to the invasive and high-risk nature of CABG in comparison

Table 3: Single/Two-Vessel Disease

Anatomical Location	CABG Recommendation	PCI Recommendation	Key Considerations
Involving proximal LAD	Class I, Level A (either modality acceptable)	Class I, Level A (either modality acceptable)	Both equally valid for prognostic benefit
Complex proximal LAD lesion	Class I, Level B (preferred for durability)	Lower priority	CABG preferred for complex LAD lesions not amenable to PCI
Not involving proximal LAD (pLAD)	Class IIb, Level C (may be considered if PCI not feasible)	Class I, Level B (in selected cases)	Debatable benefits from revascularisation, however, PCI typically first choice for non-pLAD single/double vessel disease

with PCI, patients would stay in hospitals longer, and the healing is slower.⁵⁹ The guidelines address this nuance: if a proximal LAD lesion (single or two-vessel disease) is complex and less amenable to PCI, then CABG is specifically recommended to improve symptoms and to reduce the need for repeat revascularisation (Class I, Level B).⁸ In other words, for a tough LAD lesion (or where multiple lesions, including the LAD, would require multiple stents), surgical bypass (typically using the left internal mammary artery graft to the LAD) is the preferred strategy because of its superior long-term patency and outcomes.⁶⁰⁻⁶² This recommendation acknowledges the well-known benefit of LIMA-LAD grafting for long-term survival in CAD compared to PCI.⁵⁸

In symptomatic CCS patients with one or two-vessel disease not involving the proximal LAD, the urgency for revascularisation is lower since the territory at risk is smaller.^{55, 63-64} The guidelines state that if such a patient remains symptomatic on medical therapy, PCI is recommended to improve symptoms (Class I, B).⁸ PCI in this scenario is typically straightforward (e.g. stenting an isolated right coronary or circumflex lesion) and provides effective angina relief. CABG is rarely warranted in single/double-vessel disease without proximal LAD involvement; the guidelines note that if the lesions are not amenable to PCI, CABG may be considered to relieve symptoms (Class IIb, C).⁸ An example might be a proximal dominant circumflex occlusion not suitable for a stent due to tortuosity – surgery could be an option, though it's a Class IIb (weak) recommendation, underscoring that medical therapy is also acceptable if revascularisation risk is high. In general, isolated non-LAD lesions are managed with PCI, if anything, and CABG is reserved for those unusual cases where PCI is not possible or has failed and symptoms are significant.

Across all these scenarios, it should be remembered that optimal medical therapy (anti-anginals, risk factor modification, etc.) remains the cornerstone of CCS management. Revascularisation is typically pursued when medicine is insufficient or when there is a clear prognostic benefit. The guidelines reinforce that lifestyle and pharmacologic therapy are fundamental, with revascularisation built on that foundation.⁸

Patients with Heart Failure or Left Ventricular Dysfunction

A special subgroup of CCS patients is those with heart failure and reduced ejection fraction (HFrEF) due to ischemic heart disease (often termed ischemic cardiomyopathy). In these patients (LVEF $\leq 35\%$), viability and perfusion of myocardium become considerations for revascularisation decisions. The 2024 ESC guidelines advise an aggressive approach to identify and treat coronary lesions in this population.⁸ Specifically, in heart failure patients (EF $\leq 35\%$) with suspected CAD, an invasive coronary angiogram (ICA) is recommended to evaluate for disease amenable to revascularisation (Class I, B).^{8, 65-68} The rationale is to look for CAD that could be surgically bypassed to improve prognosis, as evidenced by trials like STICH.⁶⁹ CABG can confer a survival benefit in patients with impaired EF by restoring blood flow to hibernating myocardium.⁷⁰⁻⁷¹ Thus, if significant CAD is found in an HFrEF patient after ICA, especially if it's multivessel, CABG is the preferred modality to attempt to improve long-term survival and heart failure outcomes after considering the risk-to-benefit ratio of the procedures (this is aligned with Class I recommendation for CABG in 3VD with LV dysfunction) (Table 4).⁸

PCI in the context of severe LV dysfunction is less studied, and the guidelines do not give it a primary recommendation for prognostic improvement; however, PCI may be used for symptom relief or if a patient cannot undergo surgery. One notable point in the guidelines is that for selected HFrEF patients undergoing a high-risk PCI (e.g. complex multivessel stenting in a low EF patient), the use of a mechanical circulatory support device (such as a percutaneous microaxial flow pump) may be considered to reduce peri-procedural risk (Class IIb, C).^{8, 72-74} This is more of a technical adjunct recommendation, reflecting the need for caution when doing PCI in very sick heart failure patients. Nonetheless, the overarching guideline message for ischemic cardiomyopathy is to consider surgical revascularisation (CABG) for eligible patients to improve survival, whereas PCI would be an alternative if surgery is contraindicated. All such patients should be managed by a multidisciplinary heart failure team as well, optimising medical therapy

Table 4: Heart Failure/ LV Dysfunction

Clinical Scenario	CABG Recommendation	PCI Recommendation	Key Considerations
HFrEF (EF ≤ 35%) with multivessel CAD-low surgical risk	Class I (preferred for survival benefit)	PCI not recommended	CABG shown to improve survival in ischemic cardiomyopathy
HFrEF (EF ≤ 35%) with multivessel CAD—high surgical risk	Not recommended	Class IIb, Level C (mechanical support may be considered)	Mechanical circulatory support for high-risk PCI

(e.g. beta-blockers, ACE inhibitors, SGLT2 inhibitors, etc.) alongside any revascularisation to decrease mortality.⁷

Conclusion

The 2024 ESC CCS guidelines provide a nuanced, evidence-based framework for choosing between PCI and CABG in patients with stable coronary disease.^{2,8} CABG is favoured for patients with left main or complex multivessel disease, especially in the presence of diabetes or reduced ejection fraction, due to its association with improved survival, fewer MI events, and more durable results.²⁴

PCI is an equal alternative in scenarios of less complex anatomy (low SYNTAX score) where outcomes have proven comparable, such as low-complexity left main or three-vessel disease in non-diabetics. For patients with limited one or two-vessel disease, PCI is usually sufficient for symptom relief, with CABG reserved for proximal LAD lesions or cases not amenable to stenting. Throughout all recommendations, there is a strong emphasis on individualised care: employing the Heart Team to integrate clinical factors (anatomy, diabetes, LV function, surgical risk) and patient preferences to arrive at the best revascularisation strategy. By adhering to these guidelines, clinicians can ensure that patients with chronic coronary syndromes receive therapy (whether PCI, CABG, or medical management) that is best aligned with their clinical profile and supported by the highest level of evidence.

Disclosure Statement

The author has no conflicts of interests to declare.

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